# Symmetric extremal configurations for the Erdős–Szekeres problem

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**Abstract**

We study planar point sets in general position that are invariant under a non-trivial isometry and contain no $k$ points in convex position, and compare their largest sizes with the Erdős–Szekeres value $2^{k-2}$. For reflections the problem reduces exactly to the Erdős–Tuza–Valtr function: the largest mirror-symmetric set has $2N(k, k/2, k)+2$ points for even $k$ and $2N(k, (k+1)/2, k)$ for odd $k$. With Baek's published theorem (the case $a = 4$ of that conjecture) this gives the values $2, 4, 8, 12, 32, 46$ for $k = 3, \ldots, 8$, those for $k = 7, 8$ being corollaries of Baek's theorem (upper bounds for $k \geq 4$ Lean-checked, with no SAT input), the bounds $128 \leq N_{\mathrm{mir}}(9) \leq 226$ and $188 \leq N_{\mathrm{mir}}(10) \leq 316$, and the equivalence of the conjectured mirror values with cases of the Erdős–Tuza–Valtr conjecture. For rotations we give the orbit-counting rule ($2^{k-2}$ fits a $(k-1)$-fold rotation if and only if $k-1$ is an odd prime, an odd base-2 Fermat pseudoprime or a power of 2). For $k = 8$ the largest sevenfold symmetric set in general position without a convex octagon has exactly 64 points and the largest sixfold symmetric one exactly 60, so $ES(8) = 65$ holds for these two symmetry types: the lower bounds are explicit sets (seven and five order types; exact computation), and the upper bounds are SAT certificates accepted by the verified checker cake_lpr, on top of a Lean-checked reduction and a reviewed, unrefereed written proof of the encoding (for the sixfold bound, all 64 cube formulas have cake_lpr verdicts bound to file hashes since a replay of 6 November). We also give a computer-assisted argument (SAT certificates accepted by cake_lpr, on top of unrefereed written proofs) that one orbit of rotated generic tiny clusters cannot reach 128 points for $k = 9$, and examples with 120, 253 and 991 points for $k = 9, 10, 12$. Centrally symmetric $2^{k-2}$-point sets exist for every $k \neq 4$; this is implied by published constructions, although no source inspected states it, and for even $k$ we give an independent proof. Every statement carries a rigor label.

## 1 Introduction

Let $ES(k)$ be the least $n$ such that every set of $n$ points in general position in the plane contains $k$ points in convex position. Erdős and Szekeres proved that $ES(k)$ exists [10], constructed $2^{k-2}$ points in general position without $k$ points in convex position [11], and conjectured $ES(k) = 2^{k-2} + 1$. The conjecture holds for $k \leq 6$: $ES(4) = 5$ (E. Klein, see [10]), $ES(5) = 9$ (Makai; first published proof by Kalbfleisch, Kalbfleisch and Stanton [14]) and $ES(6) = 17$ (Szekeres and Peters [23]). It is open for $k = 7$; for $k = 8$ the refereed upper bound we use is $ES(8) \leq 408$ [19].

**Symmetric search spaces.** Imposing a symmetry shrinks the search space, and optimality may be easier to prove for structured sets. Subercaseaux, Mackey, Qian and Heule (SMQH) [22] built rotational symmetry into SAT encodings and found symmetric extremal sets for $k = 6$, a fourfold one and a fivefold one with centre [22, App. C, Tables 1–2] (also [17, Figs. 3.28, 3.33]). They wrote [22, §7]: "for the Erdős–Szekeres problem, we suspect that there are symmetric constructions for all $k \geq 7$, and proving this could represent progress toward the conjectured bound $g(k) = 2^{k-2} + 1$, since it might be easier to prove optimality for more structured solutions." They name no symmetry type or order. This note collects what we can say for each type.

**The case $k = 7$.** A companion study [24, `WRITEUP/SYMMETRIC.md`] shows, with computer-assisted proofs, that every set of at least 33 points in general position mapped into itself by an isometry or affine bijection other than the identity contains a convex heptagon: the reduction to five case classes is Lean-checked, the reflection classes are Lean-checked consequences of Baek's theorem (Corollary 2.10), and the other classes rest on SAT case splits, written soundness proofs of the encodings, certificates accepted by cake_lpr and, for most threefold and half-turn pieces, verdicts recorded before the proofs were deleted. The fivefold case also follows from a computation reported earlier by Dumitru [8, §7, Table 1] (no proof files public), which has priority. For $k = 7$ only the half-turn and reflections reach $2^{k-2}$ (Table 2); for $k = 8$ sevenfold symmetric sets reach it too (Theorem 3.8), while reflections stop at 46.

Sections 2–4 treat reflections, rotations and the half-turn; Section 5 summarizes rigor, refuted statements and open problems, and Section 6 gives credit and the sources inspected.

### 1.1 Conventions and rigor labels

*General position* means that no three points are collinear; a set is in *convex position* if no point lies in the convex hull of the others; a *convex $k$-gon* of $X$ is a $k$-subset in convex position, and $\mathrm{mc}(X)$ is the largest size of one. Let $\mathrm{orient}(p,q,r) = (x_q - x_p)(y_r - y_p) - (y_q - y_p)(x_r - x_p)$. A set is *$x$-distinct* if its points have distinct $x$-coordinates. A sequence $p_1, \dots, p_m$ with increasing $x$ is an *$m$-cap* if $\mathrm{orient}(p_{i-1}, p_i, p_{i+1}) < 0$ for $1 < i < m$ (slopes decrease) and an *$m$-cup* if these are $> 0$; sequences of at most two points are both.

**Lemma 1.1** (cap–cup decomposition; standard, published [10])**.** *Let $P$ be $x$-distinct and in general position, $|P| \geq 3$, with leftmost point $a$ and rightmost point $b$. Then $P$ is in convex position if and only if the points of $P$ on or above the line $ab$ form a cap from $a$ to $b$ and those on or below it form a cup from $a$ to $b$; then $|P|$ is the sum of their sizes minus 2.*

**Rigor labels.** *Lean-checked*: compiled with Lean 4.24.0 and the repository's Mathlib, depending only on `propext`, `Classical.choice`, `Quot.sound`. *Certificate*: an unsatisfiability claim whose LRAT proof (produced by CaDiCaL 3.0.1, or 2.1.3 for the re-checks of 5 November and the sixfold replay of 6 November) was accepted by the CakeML-verified checker `cake_lpr`, which prints `s VERIFIED UNSAT`; the soundness of the encoding is a written proof. Acceptance by `lrat-check` alone is not a certificate: the `lrat-check` of drat-trim at commit `2e3b2dc` accepts an invalid proof of a satisfiable formula unless patched, a publicly known defect [7] (our record: `LRATCHECK20261105`); `lrat-check` verdicts cited here are supporting only. *Exact computation*: exact integer or algebraic arithmetic on explicit data, by at least two independently written programs unless stated otherwise. *Written proof (not refereed)*: a complete written proof, here or in the supplementary material, checked by at least one independent internal review, not refereed by a journal. *Published*: a refereed published proof. *Computational evidence* (tests and failed searches, never used as proofs), *conjecture*, and *refuted* (by an explicit, exactly checked counterexample).
**Supplementary material.** The directories `MIRRORETV`, `SYMLEAN/mirror`, `ES8SYM`, `PRIMEFOLD`, `SYMEVEN`, `SYM2`, `C6UPPER`, `C7UPPER`, `LAYERLEAN`, `SIX60` and the file `WRITEUP/SYMMETRIC.md` of the project repository [24] contain the full proofs, Lean files, point sets with SHA-256 sums, SAT encodings, logs and reproduction commands.

## 2 Mirror symmetry

### 2.1 The Erdős–Tuza–Valtr function

**Definition 2.1.** For $n \geq 3$ and $a, b \geq 2$, $N(n,a,b)$ is the largest size of an $x$-distinct set in general position with no $n$ points in convex position, no $a$-cap and no $b$-cup (Baek's notation [2, Def. 1.3]); by the cups–caps theorem $N(n,a,b) \leq \binom{a+b-4}{a-2}$ [10]. Put $E(n,a,b) = \sum_{i=n-b}^{a-2} \binom{n-2}{i}$. The *Erdős–Tuza–Valtr (ETV) conjecture* [12] states $N(n,a,b) = E(n,a,b)$ for $2 \leq a, b \leq n \leq a+b-2$; $P(n,a,b)$ denotes this statement for one triple, and $P(n,n,n)$ is the Erdős–Szekeres conjecture for $n$. Let $N_{\mathrm{mir}}(k)$ be the largest size of a set in general position, invariant under the reflection in some line, with no $k$ points in convex position. Put $L(k) = 2^{k-2}$ for odd $k$ and $L(k) = 2^{k-2} - \binom{k-2}{k/2-1} + 2$ for even $k$.

As an $n$-cup is a convex $n$-gon, $N(n,a,n)$ is the largest size of an $x$-distinct set in general position with no $a$-cap and no convex $n$-gon. One has $E(2j,j,2j) = \left(2^{2j-2} - \binom{2j-2}{j-1}\right)/2 = 1, 5, 22, 93, 386$ and $E(2j-1,j,2j-1) = 2^{2j-4} = 1, 4, 16, 64, 256$ for $j = 2, \dots, 6$, and $L(k) = 2, 4, 8, 12, 32, 46, 128, 188, 512, 774, 2048, 3174$ for $k = 3, \dots, 14$.

**Theorem 2.2** (Baek [2, Thm. 1.6]; published)**.** *For every $n \geq 3$, $N(n,4,n) = \binom{n-1}{2} + 1$: every $x$-distinct set of $\binom{n-1}{2} + 2$ points in general position contains a 4-cap or $n$ points in convex position.*

Baek formalized his theorem in Lean 4; an adaptation underlies the Lean-checked bounds below. Balko and Valtr [5] had reported $N(7,4,7) = 16$ and $N(8,4,8) = 22$ (their $N(4,7,7)$, $N(4,8,8)$) from SAT computations.

**Theorem 2.3** (Erdős–Szekeres [11], Erdős–Tuza–Valtr [12]; published)**.** *$N(n,a,b) \geq E(n,a,b)$ whenever $2 \leq a, b \leq n \leq a+b-2$.*

It uses blocks $T_i = G(n-i, i+2)$, $n-b \leq i \leq a-2$, where $G(c,d)$ is a cups–caps set of $\binom{c+d-4}{c-2}$ points with no $c$-cup and no $d$-cap (proof also in `MIRRORETV/PROOFS.md` §6). *Known cases of the ETV conjecture* in the sources inspected: $n \geq a+b-3$ (cups–caps theorem); $a = 4$ or $b = 4$ (Theorem 2.2; the range then forces $b \geq n-2$, resp. $a \geq n-2$); $n \leq 6$ (from $ES(n) = 2^{n-2} + 1$ for $n \leq 6$ and [2, Thm. 1.5]; for $n = 6$ also [5]). Schroeder reports $N(7,5,6) = 25$ (unreviewed) [21].

## 2.2 The mirror lemma

Put the mirror on the $x$-axis and write $\mu(x, y) = (x, -y)$.

**Lemma 2.4** (mirror lemma; Lean-checked)**.** *Let $C = (c_1, \dots, c_m)$ be a cap with $y(c_1), y(c_m) \geq 0$ (if $m = 1$, $y(c_1) > 0$). Then $C \cup \mu(C)$ is in convex position and has $2m - e$ points, where $e$ is the number of end points of $C$ on the axis.*

*Proof.* Let $m \geq 2$ and let $f$ be the piecewise-linear interpolation of $C$ on $I = [x(c_1), x(c_m)]$. It is concave and non-negative at both ends, hence $f \geq 0$, and interior points of $C$ lie strictly above the axis. The region $Q = \{(x, y) : x \in I,\ |y| \leq f(x)\}$ is convex (a strip intersected with the hypograph of $f$ and the epigraph of $-f$) and contains $C \cup \mu(C)$. The end points $c_1, c_m$ are extreme points of $Q$, being the highest points of $Q$ on the boundary lines of the strip. An interior $c_i$ is extreme because $f$ has a strict slope decrease at $x(c_i)$: a line through $c_i$ with slope strictly between the two slopes of $f$ there meets $Q$ only in $c_i$. By symmetry the points of $\mu(C)$ are extreme too, and a finite set of extreme points of a convex set is in convex position. The $2m$ points are distinct except that $c = \mu(c)$ for an end point on the axis. □

*Remark* 2.5 (simplicity). Checking that the closed polygon $c_1 \cdots c_m\, \mu(c_m) \cdots \mu(c_1)$ turns to the same side at every vertex does not suffice: the closed pentagon $(-18, -62), (1, 66), (-88, -82), (37, -76), (-7, 49)$ turns left at every vertex, winds twice, and its vertex $(-18, -62)$ lies strictly inside the triangle $(-88, -82), (37, -76), (1, 66)$ (exact check). A closed polygon turning to one side at every vertex is convex if it is also simple. Here it is simple because its two chains are the graphs of $f \geq 0$ and $-f \leq 0$ over one interval; the region argument avoids winding numbers. (An earlier project write-up used turn signs only; the lemma is true.)

## 2.3 The reduction and its converse

**Theorem 2.6** (upper bounds; Lean-checked)**.** *For every $j \geq 2$: $N_{\mathrm{mir}}(2j) \leq 2N(2j, j, 2j) + 2$ and $N_{\mathrm{mir}}(2j - 1) \leq 2N(2j - 1, j, 2j - 1)$.*

*Proof.* Let $S$ be symmetric in the $x$-axis, in general position, without $k$ points in convex position; put $R = S \cap \{y > 0\}$ and $A = S \cap \{y = 0\}$. Then $|S| = 2|R| + |A|$ and $|A| \leq 2$. $R \cup A$ is $x$-distinct: two points $p \neq q$ of $R$ with equal abscissa give the collinear triple $p, q, \mu(p)$, and $p \in R$, $a \in A$ give $p, a, \mu(p)$. By Lemma 2.4, $R$ has no $j$-cap, which would give $2j \geq k$ points in convex position. For $k = 2j$, $R$ has no $j$-cap and no convex $2j$-gon, so $|R| \leq N(2j, j, 2j)$ and $|S| \leq 2N(2j, j, 2j) + 2$. For $k = 2j - 1$ and $A = \emptyset$, $|S| = 2|R| \leq 2N(2j - 1, j, 2j - 1)$. Otherwise take $a \in A$ and $X = R \cup \{a\}$. A $j$-cap of $X$ would contain $a$, not as an interior point (it lies strictly below the chord of its neighbours, which lie above the axis), so as an end point, and Lemma 2.4 would give $2j - 1 = k$ points of $S$ in convex position. Hence $|R| + 1 \leq N(2j - 1, j, 2j - 1)$ and $|S| \leq 2|R| + 2 \leq 2N(2j - 1, j, 2j - 1)$. □

*Formalization.* `MIRRORETV/lean/MirrorMain.lean` proves Theorem 2.6 for any mirror line, with $N$ as a hypothesis (`MirrorOdd.mirror_even_general_line`, and `MirrorOdd.mirror_odd_general_line` for odd $k \geq 5$); `MIRRORETV/review` reformalizes it for all $j \geq 2$ (`MirrorA.theoremA_even_line`, `MirrorA.theoremA_odd_line`) and Lemma 2.4, importing only Mathlib.

**Theorem 2.7** (far-flat construction; written proof (not refereed), orientation estimates Lean-checked)**.** *Let $j \geq 2$ and let $X$ be $x$-distinct, in general position, with no $j$-cap. (i) If $X$ has no convex $2j$-gon, there is a mirror-symmetric set of $2|X| + 2$ points in general position without a convex $2j$-gon. (ii) If $X$ has no convex $(2j - 1)$-gon, there is one of $2|X|$ points without a convex $(2j - 1)$-gon.*

*Construction.* Translate $X$ so that $\min x = \min y = 0$; let $W = \max x$, $h = \max y$ and $g$ the least gap between consecutive abscissae ($g = 1$ if $|X| = 1$). In case (i) put $D = W + 2$ and $A = \{(W + 1, 0), (W + 2, 0)\}$; in case (ii) $D = \max(W, 1)$ and $A = \emptyset$. Choose $H$ with $Hg > hD$, and put $R = X + (0, H)$ and $S = R \cup \mu(R) \cup A$.

**Lemma 2.8** (cap lemma; written proof (not refereed), estimates Lean-checked)**.** *If $P \subseteq S$ is in convex position and $P \not\subseteq R$, then $P \cap R$ is a cap.*

*Proof.* Otherwise, by Lemma 1.1, $P \cap R$ has a leftmost point $u$, a rightmost point $w$ and a point $q$ strictly below the line $uw$, with $x_u + g \leq x_q \leq x_w - g$. Take $z \in P \setminus R$; then $y_z \leq 0$ and $|x_z - x_p| \leq D$ for $p \in R$. Hence $\mathrm{orient}(w, z, q) = (x_z - x_w)(y_q - y_w) + (y_w - y_z)(x_q - x_w) \leq Dh - Hg < 0$, $\mathrm{orient}(z, u, q) = (x_q - x_u)(y_z - y_u) - (y_q - y_u)(x_z - x_u) \leq -gH + hD < 0$ and $\mathrm{orient}(u, w, q) < 0$. These three numbers sum to $\mathrm{orient}(u, w, z)$, so $q$ lies inside the triangle $uwz$, a contradiction. □

*Proof of Theorem 2.7 (outline;* `MIRRORETV/PROOFS.md` *§5).* General position follows from $Hg > hD$. Let $P \subseteq S$ be in convex position. If $P$ lies in $R$ or in $\mu(R)$, then $|P| \le \mathrm{mc}(X)$. Otherwise $P \cap R$ and $\mu(P \cap \mu(R))$ are caps of $R$ (Lemma 2.8 for $P$ and $\mu(P)$), so $|P| \le 2(j-1) + |P \cap A|$; this proves (ii). In case (i), $|P| \le 2j-1$ unless $A \subseteq P$; then either $P$ misses $R$ or $\mu(R)$ and $|P| \le (j-1)+2 \le 2j-1$, or it contains $p \in R$ and $q \in \mu(R)$, the segment $pq$ crosses the axis left of $(W+1,0)$, and $(W+1,0)$ lies inside the triangle $p\,q\,(W+2,0)$. In fact $\mathrm{mc}(S) = \max(\mathrm{mc}(X),\, 2\,\mathrm{cap}(X) + [A \neq \emptyset])$, where $\mathrm{cap}(X)$ is the largest size of a cap of $X$. The orientation bounds and the barycentric step are formalized in `MIRRORETV/lean/FarFlat.lean` and `MIRRORETV/review/FarFlatExtra.lean`; the combinatorial steps are not. □

**Corollary 2.9** (exact reduction; upper bounds Lean-checked, lower bounds written proof (not refereed))**.** *For every $j \ge 2$: $N_{\mathrm{mir}}(2j) = 2N(2j, j, 2j) + 2$ and $N_{\mathrm{mir}}(2j-1) = 2N(2j-1, j, 2j-1)$.*

## 2.4 Consequences

**Corollary 2.10** (exact values for $k \le 8$)**.** $N_{\mathrm{mir}}(k) = 2, 4, 8, 12, 32, 46$ *for* $k = 3, 4, 5, 6, 7, 8$. Status: *the upper bounds for $k = 4, \dots, 8$ are Lean-checked for every mirror line, with no SAT input (for $k = 7, 8$ through the repository's formalization of Theorem 2.2); $k = 3$ is elementary; the lower bounds are exact computations on explicit sets and follow from Theorem 2.7.*

*Proof.* $N(3,2,3) = N(4,2,4) = 1$ (a 2-cap is any pair); $N(n,3,n) = n-1$, since a set with no 3-cap is a cup and an $n$-cup is a convex $n$-gon; $N(7,4,7) = 16$ and $N(8,4,8) = 22$ by Theorem 2.2. Apply Corollary 2.9. □

*Credit.* The values for $k = 7, 8$ are corollaries of Baek's theorem through the elementary Theorem 2.6; $k = 4, 5$ also follow from $ES(4) = 5$ and $ES(5) = 9$. The Lean proofs (`MirrorOdd.mirror_seven_line`, `MirrorOdd.mirror_eight_line`; `SYMLEAN/mirror`) use the repository's adaptation of Baek's formalization and bridge lemmas adapted from Schroeder's archive [21]. Also, $N(8,4,8) \le 22$ is certified by two independently written 23-point SAT encodings, each accepted by cake_lpr (`ES8SYM` §4), and $N(7,4,7) \le 16$ by 17-point SAT encodings (`MIRRORETV` §4): two of the four were certified by cake_lpr on 5 November (`LRATCHECK20261105/recheck`); the other two were accepted by lrat-check only and have no committed generator.

**Proposition 2.11** (ETV equivalence; written proof (not refereed))**.** *For $j \ge 2$: $N_{\mathrm{mir}}(2j) = L(2j)$ if and only if $P(2j, j, 2j)$ holds, and $N_{\mathrm{mir}}(2j-1) = 2^{2j-3}$ if and only if $P(2j-1, j, 2j-1)$ holds. Both hold if $ES(k) = 2^{k-2}+1$ for the corresponding $k$.*

*Proof.* The equivalences follow from Corollary 2.9, Theorem 2.3 and the values of $E$. For the last sentence: Erdős, Tuza and Valtr showed that $ES(n) = 2^{n-2}+1$ implies $N(n,a,b) \le E(n,a,b)$ for all $a, b$ with $2 \le a, b \le n \le a+b-2$ [2, Thm. 1.5] (reproved in `MIRRORETV/PROOFS.md`, Prop. 7.1). For odd $k$ it also follows from $N_{\mathrm{mir}}(k) \le ES(k) - 1$. □

So the conjectured mirror values are exactly the ETV statements $P(k, \lceil k/2 \rceil, k)$; nothing mirror-specific remains. They are known for $k \le 8$ ($k = 7, 8$ by Baek); in the sources inspected they are open from $k = 9$ on.

**Lemma 2.12** (Pascal step; written proof (not refereed), the argument of [10])**.** *For $a, b \ge 3$: $N(n,a,b) \le N(n,a-1,b) + N(n,a,b-1)$.*

*Proof.* Let $X$ be extremal and $E \subseteq X$ the set of last points of $(a-1)$-caps. $X \setminus E$ has no $(a-1)$-cap. If $c_1 \cdots c_{b-1}$ were a cup in $E$, then $c_1$ would end an $(a-1)$-cap $d_1 \cdots d_{a-2} c_1$, and either $d_{a-2} c_1 c_2$ is a cap triple (an $a$-cap) or $d_{a-2} c_1 \cdots c_{b-1}$ is a $b$-cup. Both parts inherit the other conditions. □

**Corollary 2.13** ($k = 9, 10$)**.** $128 \le N_{\mathrm{mir}}(9) \le 226$ *and* $188 \le N_{\mathrm{mir}}(10) \le 316$. Status: *the upper bound for $k = 9$ is Lean-checked (through the formalization of Theorem 2.2 at $n = 9$;* `C12Check.mirror_nine_line`*); for $k = 10$ it is a written proof (not refereed) resting on Theorem 2.2; the lower bounds are written proofs (not refereed), realized by explicit sets (exact computation).*

*Proof.* By Lemma 2.12, Theorem 2.2 and the cups–caps bound, $N(9,5,9) \le 29 + \binom{9}{3} = 113$ and $N(10,5,10) \le 37 + \binom{10}{3} = 157$; apply Corollary 2.9. The lower bounds are $2 \cdot 64$ and $2 \cdot 93 + 2$ (Theorems 2.3 and 2.7). Without Baek's theorem the cups–caps bound gives 240 and 332. □

### 2.5 Constructions for every $k$

**Theorem 2.14** (written proof (not refereed); explicit sets for $k \le 14$: exact computation)**.** *For every $k \ge 3$, $N_{\mathrm{mir}}(k) \ge L(k)$, realized with no point on the axis for odd $k$ and two for even $k$.*

*Proof.* Apply Theorem 2.7 to the set of Theorem 2.3 with $(n, a, b) = (k, \lceil k/2 \rceil, k)$. □

The sets for $k = 4, \dots, 14$ (`MIRRORETV/sets`; $k = 3$ is trivial) have largest convex subset $k - 1$, by two or three independently written exact programs (for $k = 13, 14$ by one exact dynamic program and an exact check of the hypotheses of the relevant lemma), and for $k \le 8$ also by brute force (none of the 260,932,815 eight-subsets of the 46-point set is convex).

*Remark* 2.15. A second construction places the Erdős–Szekeres blocks $T_i = G(k - i, i + 2)$ so that $\mu$ maps block $i$ to block $k - 2 - i$, the self-paired middle block (even $k$) becoming two axis points (`MIRRORETV/PROOFS.md` §9). For odd $k$ it is an instance of Valtr's construction for an arbitrary skeleton as described by Baek and Balko [3, §4] (stated there without proof, citing [11, 12]), with a mirror-symmetric skeleton; they do not state the mirror case. For $k = 7$ a 32-point mirror set also follows from the classical construction in Lovász's form [18, Thm. 2.6], with mirrored clusters (`WRITEUP/SYMMETRIC.md`). By Theorem 2.14, for odd $k$ reflections reach $2^{k-2}$; for even $k$ they fall short by $\binom{k-2}{k/2-1} - 2$ if and only if $P(k, k/2, k)$ holds (unconditionally $12 < 16$ for $k = 6$ and $46 < 64$ for $k = 8$).

## 3 Rotational symmetry

Let $\rho$ be a rotation of order $m$ about $O$; orbits of points other than $O$ are regular $m$-gons.

**Lemma 3.1** (layer lemma; Lean-checked (`LAYERLEAN`))**.** *Let $m \ge 3$, $K \le 2m$, and let $S$ be finite, in general position, $\rho$-invariant, with no $K$ points in convex position. Then $S \setminus \{O\}$ is the union of regular $m$-gons $L_1, \dots, L_t$ centred at $O$ (its convex layers), each strictly inside the convex hull of the previous one. If $m$ is even and $S \ne \{O\}$, then $O \notin S$.*

*Proof.* The vertex set of $\mathrm{conv}(S \setminus \{O\})$ is $\rho$-invariant, hence a union of orbits of $m$ points; two orbits would be $2m \ge K$ points in convex position, so it is one orbit. The other points lie strictly inside its hull (not on an edge, by general position); repeat. For even $m$, $\rho^{m/2}$ is the half-turn and $O, p, -p$ are collinear. □

SMQH [22, §3.2] use the special case of 16 points, fourfold symmetry and no convex hexagon (four layers of four points).

**Proposition 3.2** (size compatibility; Lean-checked core)**.** *A finite $\rho$-invariant set has $mt + [O \in S]$ points. Hence an $m$-fold symmetric set of $2^{m-1}$ points ($m \ge 2$) can exist only if $m$ is a power of 2 (and $O \notin S$) or $m$ is an odd prime or an odd base-2 Fermat pseudoprime (and $O \in S$); for these $m$ such sets in general position exist. So $2^{k-2}$ is a possible size for $(k-1)$-fold symmetry exactly for $k = 3, 4, 5, 6, 8, 9, 12, 14, 17, 18, 20, 24, \dots$ and $k = 342, 562, 646, \dots$ (from $m = 341, 561, 645$).*

*Proof.* Only $O$ is fixed by $\rho^i$, $0 < i < m$. If $m$ is even, $mt$ and $2^{m-1}$ are even, so $O \notin S$ and $m \mid 2^{m-1}$, i.e. $m = 2^a$. If $m > 1$ is odd, $m \nmid 2^{m-1}$, so $O \in S$ and $2^{m-1} \equiv 1 \pmod m$: $m$ is prime or, by definition, a base-2 pseudoprime; every odd prime qualifies by Fermat's little theorem. Conversely take orbits with distinct radii and generic phases (and $O$ if needed). □

*Status.* Lean-checked (`PRIMEFOLD/compat/lean`): the orbit count, the classification of $m$ with $2^{m-1} \equiv 0, 1 \pmod m$, and that $|S| = 2^{m-1}$ forces the stated alternative. Not formalized here: the step that every odd prime is compatible (Fermat's little theorem) and the genericity step (written proof, not refereed). Antecedents: Fermat's little theorem; SMQH [22, §6.2] apply the same count to 32 points, and Mackey's 16-point layer formations [17] are the same count. In particular $k = 7$ is not compatible ($32 \equiv 2 \pmod 6$).

### 3.1 Small $k$

**Proposition 3.3** ($k = 5, 6$)**.** *(a)* Exact computation. *Eppstein's 8-point pentagon-free figure [9, Fig. 1.2] (two nested squares) is fourfold symmetric and in general position: the book's circle centres are exactly symmetric (up to print rounding), and the 2007 SVG version becomes so when 342.002 is replaced by 342, with the same orientations. (b)* Exact computation and written proof (not refereed). *Up to reflection there is exactly one order type of fourfold symmetric pentagon-free 8-point sets in general position: with squares $\{1, i, -1, -i\}$ and $\{i^\ell b\}$, $b = x + iy$, $x, y > 0$,*

$x+y<1$, *the set qualifies if and only if* $x^2+y^2<|x-y|$*; e.g.* $\{(\pm10,0),(0,\pm10),(3,1),(-1,3),(-3,-1),(1,-3)\}$. *(c)* Exact computation. *The fivefold* 16*-point set of [22, Table 2] (orbits of* $(-12,-17)$*,* $(-15,2)$*,* $(-13,0)$ *and the centre) is in general position without a convex hexagon.*

Both sizes are maximal by $ES(5)=9$ and $ES(6)=17$. The texts inspected do not call Eppstein's figure symmetric; (b) may be implicit in the order-type database of [1], which we did not consult.

## 3.2 Rotated clusters and their limit at $k=9$

For a finite set $C$ and an open arc $I$ of directions, $f_C(I)$ is the largest number of points of a chain of $C$ whose edge directions lie in $I$ and increase counter-clockwise.

**Lemma 3.4** (blow-up formula for tiny clusters; written proof (not refereed))**.** *Let* $B$ *be finite in general position, let* $C_b$ *(*$b\in B$*) be finite sets in general position, and assume that no segment of any* $C_b$ *is parallel to a line through two points of* $B$*. For small* $\varepsilon>0$ *the set* $S_\varepsilon=\bigcup_b(b+\varepsilon C_b)$ *is in general position and* $\mathrm{mc}(S_\varepsilon)=\max\bigl(\max_b \mathrm{mc}(C_b),\ \max_{B'}\sum_{b\in B'} f_{C_b}(I_b(B'))\bigr)$*, where* $B'$ *runs over subsets of* $B$ *in convex position with* $|B'|\ge 2$ *and* $I_b(B')$ *is the open arc of directions from* $\overrightarrow{b^-b}$ *counter-clockwise to* $\overrightarrow{bb^+}$*,* $b^\pm$ *being the neighbours of* $b$ *in* $B'$*.*

This is the tiny-cluster form, with rotated cluster directions, of the blow-up method of Erdős and Szekeres [11], Valtr, and Baek and Balko [3]. Without the parallelism hypothesis it is false ($B=\{(0,0),(1,0)\}$, $C_1=\{(0,0),(1,0)\}$, $C_2=\{(0,1)\}$). The *rotated-cluster family* replaces each vertex of a regular $m$-gon by the same tiny cluster $C$, rotated with the vertex, and adds the centre for odd $m$. Measure directions in the frame of a vertex from the counter-clockwise tangent, in units of $\pi/m$, and let $g(x)$, $h(x)$ be the longest counter-clockwise chains of $C$ with directions in $(0,x)$ and in $(-x,0)$.

**Proposition 3.5** (one orbit; written proof (not refereed))**.** *For a generic cluster* $C$ *(as in Lemma 3.4) and small* $\varepsilon$*, the rotated-cluster set has no* $m+1$ *points in convex position if and only if (i)* $g(x)+h(x)\le x+1$ *for* $x=1,\dots,m-1$*, (ii) a condition involving the centre holds (odd* $m$ *only;* `PRIMEFOLD/cluster/PROOFS.md`*), and (iii)* $\mathrm{mc}(C)\le m$*.*

**Theorem 3.6** (limit of the family; certificate (cake_lpr), on top of Lemma 3.4, Proposition 3.5(i) and the soundness of the encoding, written proofs (not refereed); lower bounds exact computation)**.** *Let* $\nu(m)$ *be the largest size of a generic cluster satisfying (i). Then* $\nu(4)=2$*,* $\nu(5)=3$*,* $\nu(6)=5$*,* $\nu(7)=9$ *and* $\nu(8)\le 14$*. Hence a rotated-cluster set with generic cluster and no* $m+1$ *points in convex position has at most* $8,16,30,64,112$ *points for* $m=4,\dots,8$*; for* $k=9$ *this family cannot reach* $2^7=128$*.*

A sound SAT relaxation of (i) is unsatisfiable for $(m,n)=(4,3),(5,4),(6,6),(7,10),(8,15)$ (two encoders, first checked with lrat-check only; on 5 November all seven formulas were regenerated with matching hashes and their proofs accepted by cake_lpr, and an independently written encoder gave 10 of 10 formulas accepted by cake_lpr; `LRATCHECK20261105/recheck`); the values for $m\le 7$ are attained by exactly checked sets. Not covered: cluster segments exactly parallel to a chord, and several orbits. The family reaches $2^{k-2}$ for $k=3,4,5,6,8$ with clusters of $1,1,2,3,9$ points, for $k=5,6$ with the order types of Proposition 3.3 ($k=5$: up to reflection; exact computation).

*Remark* 3.7 (explicit and flat clusters). A max/min recursion gives clusters of $B(h)=1,3,9,28,90,297,1001$ points for $h=1,\dots,7$ ($m=2h+1$; no log kept for $h=7$), realized exactly for $h=2,\dots,5$ ($16,64,253,991$ points). For these $h$, $B(h)=\binom{2h}{h-1}-\binom{2h}{h-2}$ (A000245 [20]), which equals $(2^{m-1}-1)/m$ for $m=3,5,7$, its floor for $m=9$, and is smaller for $m=11,13,15$ (computational evidence; maximality of $B(h)$ is a conjecture). One orbit of *flat* clusters with generic flattening directions has fewer than $2^{m-1}$ points for $m\ge 7$ (written proof (not refereed)); without genericity it fails: the last set of Table 1 is one orbit of chord-aligned flat 9-point clusters plus the centre.

## 3.3 Sevenfold symmetric 64-point sets

**Theorem 3.8** (exact computation)**.** *The seven sets of Table 1 are sevenfold symmetric sets of* 64 *points in general position (the centre and nine orbits, hence nine nested regular heptagons) with no* 8 *points in convex position, and they are pairwise non-equivalent order types.*

The coordinates are integer combinations of powers of $\zeta_7$, so the symmetry is exact; all 41,664 orientations are nonzero (exact, and again with 110-digit decimals), and convex $k$-gons were counted by two programs. The numbers of convex 4-gons differ and are invariant under relabelling and reflection.

A subset $C\subseteq S$ is a *line-module* of $S$ if no line through two points of $S\setminus C$ separates two points of $C$; it is trivial if $|C|\le 1$ or $|C|\ge|S|-1$. This depends only on the orientations, so the multiset of module sizes is an order-type invariant, also under reflection.

| set (PRIMEFOLD/sets/k8) | origin | convex 4-gons | convex 7-gons | line-modules |
|---|---|---|---|---|
| c7_64pts_small* | annealing of radii and phases | 416,052 | 4,688,763 | none of size 2–60 |
| c7_64pts_9layers_centre* | same, other rounding scale | 416,220 | 4,691,381 | none of size 2–60 |
| c7_64_own_s1000 | second annealer | 392,966 | 3,328,145 | none of size 2–20 |
| c7_64pts_9layers_centre_own* | second annealer, other scale | 392,553 | 3,252,475 | none of size 2–56 |
| p7_compound_64* | two-level 9-point clusters | 470,764 | 18,396,499 | 91 of size 2–20, seven of size 9 |
| rule_p7_64pts* | cluster rule, $h = 3$ | 487,711 | 30,031,745 | 91 of size 2–20, seven of size 9 |
| m7_x2_G9s_centre | flat 9-point clusters | 462,581 | 14,719,287 | 77 of size 2–20, seven of size 9 |

Table 1: Sevenfold symmetric 64-point sets without a convex octagon (exact computation); *: also in ES8SYM/sets. For comparison, the classical Erdős–Szekeres 64-point set has 422,186 convex 4-gons and the centrally symmetric set of Section 4 has 345,986.

**Lemma 3.9** (line-modules; written proof (not refereed)). *(a) Every cluster of an $(X, Y)$-blow-up [3, Def. 13] of a base with at least two points is a line-module. (b) In a blow-up by sufficiently small generic clusters as in Lemma 3.4, every cluster is a line-module.*

(For (a): lines through two other clusters see all of $C$ like the corresponding base point, and the almost-vertical condition handles lines inside one cluster; ES8SYM/proofs/structure.md.)

**Proposition 3.10** (exact computation, with Lemma 3.9). *(a) For each starred set of Table 1, the only permutations preserving all orientations are the seven rotations, and none reverses all orientations. The classical Erdős–Szekeres 64-point set has no non-trivial automorphism, so no set with a rotation or reflection symmetry is order-type equivalent to it. (b)* c7_64pts_small *and* c7_64pts_9layers_centre *have no line-module with 2 to 60 points (their only non-trivial modules are the complements of two or three consecutive hull vertices), and* c7_64pts_9layers_centre_own *has none with 2 to 56. So none of them is order-type equivalent to a blow-up as in Lemma 3.9 with a cluster of that size; in particular not to the classical construction (modules of 6, 15, 20 points), to Valtr's construction in the form of [3, §6.1] (clusters of $\binom{6}{i}$ points) or to the sets of [3, Thm. 19] for $k = 8$ (clusters of 7 or 22 points).*

The last three sets of Table 1 are seven 9-point clusters around a heptagon plus the centre; whether they are equivalent to some Baek–Balko blow-up was not checked.

## 3.4 Records for $k = 9, 10, 12$

**Theorem 3.11** (exact computation). *There are an eightfold symmetric set of 120 points (15 nested regular octagons) without 9 points in convex position, a ninefold symmetric set of 253 points (28 layers and the centre) without 10, and an elevenfold symmetric set of 991 points (90 layers and the centre) without 12 (*PRIMEFOLD/sets*).*

All orientations (280,840, 2,667,126, 161,716,335) are exactly nonzero, and largest convex subsets were computed by two or three programs. The 253- and 991-point sets come from the rule of Remark 3.7. As ninefold sizes are $9t$ or $9t + 1$, 253 is the largest possible ninefold size if $ES(10) = 257$. By Theorem 3.6 the 120-point set is not one orbit of generic tiny clusters; no eightfold set found (104, 112, 120 points) has a line-module with 2 to $n - 5$ points. Searches for one more layer failed (computational evidence only).

## 3.5 The case $k = 8$

**Proposition 3.12** ($k = 8$; (a) and the arithmetic of (b) Lean-checked, the rest written proof (not refereed)). *Let $S$ be $m$-fold symmetric, in general position, without 8 points in convex position. (a) If $m \geq 8$, then $S \subseteq \{O\}$. (b) If $m \in \{4, 5, 6, 7\}$, then $S \setminus \{O\}$ consists of $t$ layers (Lemma 3.1), and $|S| \geq 65$ needs $t \geq 17, 13, 11, 10$ respectively.*

Part (a): an orbit is a convex $m$-gon (RefuteC6.part_a). The largest layer counts found are 13, 11 (with centre), 10, 9 (with centre). Since threefold sizes are $3t$ or $3t + 1$ and $3 \cdot 21 + 1 < 65$, a threefold counterexample to $ES(8) = 65$ exists if and only if some 22-orbit threefold set without centre is octagon-free. For $m = 6, 7$ the counts 10 and 9 are the largest possible: every configuration of 11 nested regular hexagons, resp. 10 nested regular heptagons, about a common centre and in general position contains 8 points in convex position (certificates accepted by cake_lpr for two separately implemented encoders, C6UPPER, C7UPPER; the reduction to these statements is Lean-checked in LAYERLEAN; the soundness of the encoding is a written proof (not refereed); what was replayed is stated in Table 3). With the explicit sets (exact computation; SIX60, Theorem 3.8) the sixfold maximum is exactly 60 and the sevenfold maximum exactly 64. For $m = 3, 4, 5$ upper bounds come only from $ES(8) \leq 408$, giving 407 (the repository's Lean-checked $ES(8) \leq 311$, not re-checked here, gives 310).

| symmetry | sizes | $k=7$: maximum | $k=8$: largest known | $k=8$: status | least size violating $ES(8)=65$ |
|---|---|---|---|---|---|
| reflection | $2t+b$, $b\le 2$ | 32 | **46** (exact value) | Cor. 2.10 (via Baek) | impossible |
| half-turn | $2t$ | 32 | 64 | exact computation$^\dagger$ | 66 |
| 3-fold | $3t$, $3t+1$ | 31 | 61 (with centre) | exact computation | 66 (no centre) |
| 4-fold | $4t$ | 28 | 52 | exact computation | 68 |
| 5-fold | $5t$, $5t+1$ | 31 | 56 (with centre) | exact computation | 65 |
| 6-fold | $6t$ | 30 | **60, exact** | exact comp.; certificate (`C6UPPER`) | 66 (excluded) |
| 7-fold | $7t$, $7t+1$ | 1 | **64, exact** (with centre) | exact comp.; certificate (`C7UPPER`) | 70 (excluded) |
| order $\ge 8$ | $0, 1$ | 1 | **1** (exact value) | Lean-checked | impossible |

Table 2: Largest symmetric sets without a convex $k$-gon. $k=7$: companion study (`WRITEUP/SYMMETRIC.md`), computer-assisted except reflections (Lean-checked via Baek) and orders $\ge 7$; the fourfold exclusion of 32 points was stated in [22, §6.2], and a certificate for it (six realizability cuts and an LRAT proof accepted by cake_lpr) is in `C4CUT`. $k=8$: sets in `ES8SYM/sets`, and for 6-fold in `SIX60/sets` (five 60-point sets of pairwise different order types); the half-turn value is 64 if $ES(8)=65$, and the 6- and 7-fold values are 60 and 64 unconditionally, with lower bounds by exact computation and upper bounds by certificates accepted by cake_lpr (`C6UPPER`, `C7UPPER`; Proposition 3.12 and Table 3); $^\dagger$also implied by published results (Theorem 4.1). $t$ counts the orbits with more than one point.

**Conjecture 3.13** (conjecture)**.** *For every $k\ge 3$ for which $2^{k-2}$ is compatible with $(k-1)$-fold symmetry (Proposition 3.2) there is a $(k-1)$-fold rotationally symmetric set of $2^{k-2}$ points in general position without $k$ points in convex position.*

It holds for $k=3,4,5,6,8$ (trivially for $k=3,4$; Proposition 3.3 and Theorem 3.8) and is open for $k=9,12,14,17,\dots$. It is a specific form of the SMQH suspicion quoted in Section 1.

# 4 Central symmetry

A centrally symmetric set in general position with at least two points does not contain its centre, so its size is even; four points $\pm p, \pm q$ in general position form a parallelogram.

**Theorem 4.1** (implied by published constructions; status by case below)**.** *For every $k\ge 3$ with $k\ne 4$ there is a set $S=-S$ of $2^{k-2}$ points in general position, $0\notin S$, with no $k$ points in convex position. For $k=4$ there is none.*

*Credit and status. Odd $k$*: Valtr's construction as described by Baek and Balko [3, Def. 13, Lemma 14, §6.1], with a centrally symmetric base and paired clusters; the companion study (directory `SYM2`) gives a cubic specialization with a direct proof (written proof (not refereed), re-derived in an internal audit) and exact coordinates, for $k=7$ the twofold 32-point set asked for in [22, §7]. $k=6$: the fourfold 16-point sets of [22, App. C, Table 1] and [17, Fig. 3.28] are centrally symmetric (exact computation); neither source says so. *Even $k\ge 8$*: implied by [3, Lemma 14] applied to either 16-point set with a half-turn-symmetric labelling that is not a threshold (Remark 4.4); a written argument from the published Lemma 14 (whose proof is omitted in the conference version and which we did not see), with exactly computed base data, not covered by the internal review of Theorem 4.3; no source inspected states this consequence. *Even $k\ge 6$, independently*: Theorem 4.3, which does not use Lemma 14.

*The construction for even $k$.* Fix even $k\ge 6$, $n=k-2$, and base abscissae $\{x_0<\dots<x_n\}=\{-(n-1),\dots,-3,-1,0,1,3,\dots,n-1\}$, so $x_{n/2}=0$ and $x_{n-j}=-x_j$; three distinct base abscissae sum to 0 only in the forced triples $\{x_j,0,-x_j\}$ (three odd numbers have an odd sum). Put $q_j=(x_j,x_j^3)$. For $j<n/2$ let $B_j=\{q_j+(\varepsilon u,\varepsilon^2 v):(u,v)\in G(j+2,k-j)\}$ and $B_{n-j}=-B_j$; let $B_{n/2}=\varepsilon^2 M_0$, where $M_0=-M_0$ is an $x$-distinct set of $\binom{n}{n/2}$ points in general position, $0\notin M_0$, with no $(k/2+1)$-cup or cap, slopes in $(-\frac12,\frac12)$ and $|v|<|u|/2$ for all its points (a small copy of $G(k/2,k/2+1)$ with its negative). Then $|S_\varepsilon|=\sum_j\binom{k-2}{j}=2^{k-2}$.

**Lemma 4.2** (blow-up lemma for the cubic base; written proof, checked by a referee-style independent review, not journal-refereed)**.** *Let $S$ be finite, in general position, partitioned into clusters $B_0,\dots,B_n$ such that: (H0) $S$ is $x$-distinct; (H1) $B_i$ lies left of $B_j$ for $i<j$; (H2) every slope between two clusters exceeds every slope inside a cluster; (H3) for $a<b<c$ with $x_a+x_b+x_c\ne 0$ and $p\in B_a$, $q\in B_b$, $r\in B_c$, $\operatorname{sign}\operatorname{orient}(p,q,r)=\operatorname{sign}(x_a+x_b+x_c)$; (H4) for $j<n/2$, $p\in B_j$, $q\in B_{n/2}$, $r\in B_{n-j}$ with $x(p)+x(r)\ne 0$, $\operatorname{sign}\operatorname{orient}(p,q,r)=-\operatorname{sign}(x(p)+x(r))$; (H5) $B_j$ has no $(j+2)$-cup and no $(k-j)$-cap. Then $S$ has no $k$ points in convex position.*

*Sketch; full proof in* `SYMEVEN/PROOF.md`. Let $P$ be convex with leftmost point in $B_l$ and rightmost point in $B_r$. If $l = r$, $|P| \le k-2$ by (H5) and Lemma 1.1. Otherwise (H2) makes the lower cup of $P$ a cup in $B_l$ followed by single points of later clusters, and the upper cap single points followed by a cap in $B_r$; with $t$ and $s$ single points, $|P| \le k-(r-l)+t+s$. By (H3) the turns at the first single point of the cup and the last single point of the cap put these single points in disjoint clusters strictly between $l$ and $r$, so $|P| \le k-1$, except when both chains pass through the middle cluster and $r = n-l$; there (H4) and (H0) leave only $P = \{u_0, q, q', w\}$. This is the only use of $k \ge 6$. □

**Theorem 4.3** (written proof, checked by a referee-style independent review (correct and complete); not journal-refereed, not machine-checked)**.** *For every even $k \ge 6$ and all sufficiently small $\varepsilon > 0$, $S_\varepsilon$ is in general position, $S_\varepsilon = -S_\varepsilon$, $0 \notin S_\varepsilon$, $|S_\varepsilon| = 2^{k-2}$, and (H0)–(H5) hold; hence $S_\varepsilon$ has no $k$ points in convex position.*

The decisive condition is (H4): $\mathrm{orient}(p,q,r) = x_j^3(x(p)+x(r)) + O(\varepsilon^2)$, so the middle cluster must be small compared with $\varepsilon$ times the least $x$-gap of the outer clusters; at the outer scale convex $k$-gons appear (exact computations for $k = 6, 8$). *Exact instances* (`SYMEVEN`): integer sets for $k = 6, 8, 10$ with 16, 64, 256 points satisfy (H0)–(H5) and general position exactly and have largest convex subsets $5, 7, 9$ (two dynamic programs each, re-checked by the review with its own code).

*Remark* 4.4 (the route through Lemma 14). By Lemma 14 of [3], an $(X, Y)$-blow-up of an $x$-sorted base $p_1, \dots, p_N$ in general position has no $k$ points in convex position if $x_i, y_i \ge 0$, $x_i + y_i \le k-1$ and $x_i + y_j \le k-1-s_{i,j}$ $(i < j)$, $s_{i,j}$ being the largest convex subset with leftmost point $p_i$ and rightmost point $p_j$. Sweep a 16-point set of [22, 17] so that $p_{17-i} = -p_i$, label positions 1–7 and 9 left and the others right, and give a left point $(x, k-1-x-s_i)$ and a right point $(k-1-x-s_i, x)$, $k = 6+2x$, where $s_i$ is the largest convex subset with $p_i$ as rightmost (left points) or leftmost point (right points). The conditions reduce to exactly checked facts about the base, the size is $2^{k-2}$, and clusters can be paired with their negatives. Explicit 64- and 256-point sets from this route have largest convex subsets 7 and 9 (exact computation); on these bases the threshold labelling of [3, Cor. 15] gives at most 62 points for $k = 8$.

*The case $k = 8$.* As sizes are even, "the central maximum is 64" is exactly the centrally symmetric case of $ES(8) = 65$, and any proof must be tight on the 64-point examples. The mirror argument does not transfer (written proofs (not refereed), `ES8SYM/proofs/central.md`): the projective map $(x, y) \mapsto (x/y, 1/y)$ turns a half-turn into a reflection but preserves convex position only inside an open half-plane; an outward chain $p_1, \dots, p_j$ $(j \ge 3$, increasing angles in a half-plane, left turns) gives $2j$ points $\pm p_i$ in convex position if and only if $\mathrm{orient}(p_{j-1}, p_j, -p_1) > 0$ and $\mathrm{orient}(-p_j, p_1, p_2) > 0$, which outwardness does not imply $(p = (10,1), (0,10), (-1,3))$; and a set with a "deep" half $H$ (the origin and $-H$ strictly on one side of every line through two points of $H$) has at most 56 points.

# 5 What is proved, and open problems

**Refuted statements** (exactly checked counterexamples). (1) A rotation-chain lemma assuming only left turns (counterexamples in the 52- and 54-point sets; corrected form: written proof (not refereed)). (2) "An outward chain with its antipodes is in convex position." (3) "The mirror bound 58 transfers to central symmetry" and (4) "the projective map preserves octagon-freeness" (the 64-point central sets). (5) "Rotations of order $\ge 3$ fall short of $2^{k-2}$, as for $k = 7$" (Theorem 3.8). (6) The flat-cluster bound without genericity. (7) The ETV inequality as printed in [18, p. 443], which would give $ES(n) \le 2^{n-2}$ (the lower bound is presumably meant).

**Open problems.** (1) $P(9,5,9)$ and $P(10,5,10)$: is $N_{\mathrm{mir}}(9) = 128$ and $N_{\mathrm{mir}}(10) = 188$? Likewise every $k \ge 9$; SAT at 65 or 94 points is far beyond the 23-point instance, and Baek's method is developed for $a = 4$. (2) Is there a mirror-symmetric heptagon-free 32-point set with two points on the axis (Theorem 2.6 allows it)? (3) Conjecture 3.13 for $k = 9$ (best 120 of 128; by Theorem 3.6 one orbit of generic tiny clusters does not suffice) and $k = 12$ (best 991 of 1024); does a generic cluster of 93 points satisfying Proposition 3.5 exist for $m = 11$? (4) Upper bounds below $ES(8) - 1$ for the 3-, 4- and 5-fold rotation types at $k = 8$. The 6- and 7-fold cases are settled (exactly 60 and 64; the upper bounds by SAT certificates on top of the reviewed, unrefereed encoder proof). Threshold runs of 5 November on the 5-fold (13 layers) and 4-fold (17 layers) formulas, with added chord-lemma clauses, closed no phase-order cube (fivefold: 19 of 24 ran for up to about 15 hours, 5 never started; fourfold: 11 of 12 ran for about 15 hours, one was lost; evidence only, `ROT45CHORD20261105`). Is some threefold 22-orbit set octagon-free? (5) Does every centrally symmetric set of 66 points in general position contain a convex octagon? (6) Formalize Theorem 2.7, the constructions and Lemma 4.2.

| statement | status |
|---|---|
| Lemma 2.4; Theorem 2.6; $N_{\mathrm{mir}}(k) \le 4, 8, 12, 32, 46$ $(k = 4, \dots, 8)$; $N_{\mathrm{mir}}(9) \le 226$ | Lean-checked, no SAT input ($k = 7, 8, 9$ via the Lean formalization of Baek's theorem) |
| Theorems 2.7, 2.14; Prop. 2.11; Lemma 2.12; $N_{\mathrm{mir}}(10) \le 316$ | written proof (not refereed); estimates of Thm. 2.7 Lean-checked |
| $N(8, 4, 8) \le 22$; $N(7, 4, 7) \le 16$ by SAT | certificates (cake_lpr), 2 encodings each; for $N(7, 4, 7)$ two more encodings lrat-check only |
| Explicit mirror sets; Props. 3.3, 3.10; Thms. 3.8, 3.11; $k = 8$ lower bounds (Table 2); central instances | exact computation |
| Lemma 3.1 | Lean-checked (`LAYERLEAN`: hull vertices one orbit, layers, strict nesting in orientation form, $O \notin S$ for even $m$) |
| Lemmas 3.4, 3.9; Prop. 3.5; Prop. 3.3(b); $k = 8$ central remarks | written proof (not refereed) |
| Proposition 3.2; Proposition 3.12 | core / part (a) Lean-checked; rest written proof |
| Theorem 3.6 | certificate (cake_lpr: 7 of 7 regenerated formulas with matching hashes, and 10 of 10 from an independent encoder), plus written proofs |
| $k = 8$, sevenfold maximum 64: upper bound (`C7UPPER`) | certificate (cake_lpr, two encoders; both proofs replayed against regenerated formulas with matching hashes on 4 November); reduction Lean-checked; encoding soundness written proof (not refereed) |
| $k = 8$, sixfold maximum 60: upper bound (`C6UPPER`) | certificate (cake_lpr, two encoders): in the hash-bound cake_lpr replay of 6 November all 64 cube formulas (32 per encoder), regenerated with matching hashes, were solved again and every fresh proof was accepted, each verdict bound to the formula and proof hashes (`C6UPPER` §6); before it, the solve-time logs of 3 November recorded cube labels only, and 6 of the 64 had a hash-bound cake_lpr replay (4 November). Reduction Lean-checked; encoding soundness and reflection unit written proof (not refereed) |
| Theorem 4.3, Lemma 4.2 | written proof, referee-style internal review |
| Theorem 4.1, odd $k$ (`SYM2`) | written proof (not refereed) |
| Theorem 4.1, even $k \ge 8$, via Lemma 14 | written argument from a published lemma, not internally reviewed |
| $B(h)$ (Remark 3.7); failed searches | computational evidence |
| Conjecture 3.13; maximality of $B(h)$; $N_{\mathrm{mir}}(k) = L(k)$, $k \ge 9$ | conjecture |

Table 3: Rigor of the main statements; details and reproduction commands are in the README of each directory.

# 6 Literature and credit

**Antecedents.** Erdős–Szekeres [10, 11]: the conjecture, the cups–caps theorem, the $2^{k-2}$-point construction. Erdős–Tuza–Valtr [12]: the ETV conjecture, its lower bound and its derivation from the Erdős–Szekeres conjecture. Baek [2]: the case $a = 4$ and its Lean formalization, which imply the mirror values for $k = 7, 8$ (reported by SAT in [5]). Baek–Balko [3]: $(X, Y)$-blow-ups, Lemma 14 and Valtr's construction, which imply the odd-$k$ mirror lower bound and, with the 16-point sets of [22, 17], the central sets. Morris–Soltan [18]: Lovász's form of the classical construction (the mirror set for $k = 7$). SMQH [22] and Mackey [17]: symmetric encodings, the $k = 6$ sets, the special case of the layer lemma, the orbit count for 32 points, the statement of the fourfold 32-point exclusion for $k = 7$, the suspicion of Section 1. Eppstein [9]: the figure for $k = 5$. Fermat: the size rule. Schroeder [21]: Lean bridge lemmas. Dumitru [8]: a reported computation implying the fivefold case for $k = 7$, with priority; no statement on symmetric sets for $k \ge 8$ was found in it.

**Sources inspected** (1–2 November 2026; logged searches in `MIRRORETV/lit`, `PRIMEFOLD/lit`; list in `SYMEVEN/README.md` §6.3). Read in full: [2] (arXiv v2), [3] (SoCG version), [22] (arXiv v1) and its code, [17], [11] (§§1–2), [18], [4], [25], [5], [23] (§5), Aichholzer et al. (EuroCG 2019). Searched: [16], [13] ($2^{k-2}$-point sets for $k \le 8$, no symmetry stated), [15] (empty hexagons in axially symmetric sets, a different property), [8], [21], [19], arXiv:2203.07518, 2312.01223, 1602.03075, 2502.02155, 1801.07238, 2505.15366, SSRN 7334758, Oberwolfach Report 2024/3, Wikipedia, MathWorld, erdosproblems.com #107, the OEIS ($4, 12, 46, 188, 774$: no entry; $1, 5, 22, 93, 386$ is A000346, with no geometric comment), and indexes (arXiv, OpenAlex, Semantic Scholar, Crossref, zbMATH Open, MathOverflow, partly Google Scholar). Not inspected: [12], the journal versions of [2, 6, 3], the CICM chapter of [22], the database of [1], several books and theses (`SYMEVEN/README.md` §6.3), paywalled databases and most sources not in English.

**Later checks for $k = 8$** (evidence only). A check on 4 November (`C6C7REVIEW/LITERATURE.md`) and searches on 6 November 2026 (arXiv listings and full texts, among them [16, 15], and the abstract and version history of [8]; Crossref for the CICM chapter of [22], whose full text was not accessed; GitHub, including the public Lean repository `GLitchfield/erdos-szekeres-convex-poly`, created 1 August 2026 and last pushed 20 October 2026, in which no file on symmetric sets was found; general web searches) found no work on symmetric sets without 8 points in convex position. SMQH [22] stop at $k = 7$.

No matching prior result was found in these sources for: the identities of Corollary 2.9, the far-flat construction, the odd-$k$ reduction with axis points, the even construction with two axis points, Proposition 2.11, and the derivation of $N_{\mathrm{mir}}(7) = 32$ and $N_{\mathrm{mir}}(8) = 46$ from Baek's theorem (the odd-$k$ lower bound $2^{k-2}$ itself is an instance of Valtr's construction, and for $k = 7$ also follows from Lovász's form); rotationally symmetric sets of order at least 3 for $k \geq 8$ (Theorems 3.8, 3.11 and the $k = 8$ lower bounds), the sixfold and sevenfold upper bounds for $k = 8$ (Proposition 3.12), Proposition 3.3(b), the size rule stated for all $k$, the rotated-cluster family with Theorem 3.6, Proposition 3.10 and Conjecture 3.13 in this form; the statement of Theorem 4.1 for even $k \geq 8$, which is nevertheless implied by [3, 22]; and the $k = 8$ remarks at the end of Section 4. The sources contain symmetric sets only for $k \leq 7$. This is not a novelty claim.

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